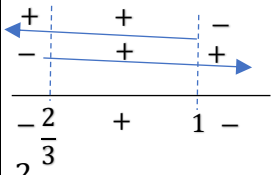
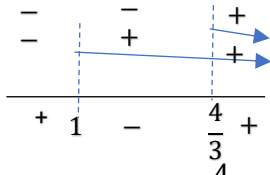
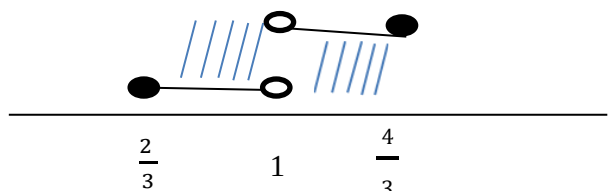


NO.	SOLUTION	MARKS	REMARKS
PART A			
1(a)	$\lim_{x \rightarrow 0} \left(\frac{\sqrt{2x+9}-3}{x} \right) \left(\frac{\sqrt{2x+9}+3}{\sqrt{2x+9}+3} \right)$ $= \lim_{x \rightarrow 0} \left(\frac{2}{\sqrt{2x+9}+3} \right)$ $= \frac{2}{\sqrt{0+9}+3}$ $= \frac{1}{3}$	B1 K1 K1 J1	Multiply by correct conjugate Simplify Substitute value
1(b)	$\lim_{x \rightarrow 1} \frac{3x^2 - 2x - 1}{ x-1 } \quad x-1 = \begin{cases} x-1, & x \geq 1 \\ -(x-1), & x < 1 \end{cases}$ $\lim_{x \rightarrow 1^+} \frac{(3x+1)(x-1)}{x-1} = \lim_{x \rightarrow 1^+} 3x+1 = 4$ $\lim_{x \rightarrow 1^-} \frac{(3x+1)(x-1)}{-(x-1)} = \lim_{x \rightarrow 1^-} -(3x+1) = -4$ Since $\lim_{x \rightarrow 1^-} \frac{3x^2 - 2x - 1}{ x-1 } \neq \lim_{x \rightarrow 1^+} \frac{3x^2 - 2x - 1}{ x-1 }$ $\therefore \lim_{x \rightarrow 1} \frac{3x^2 - 2x - 1}{ x-1 } \text{ does not exist.}$	K1 J1 J1 K1 J1	Factorize and define modulus correctly Get correct limit fr the right Get correct limit fr the left Conclusion
Total		9 marks	

NO.	SOLUTION	MARKS	REMARKS
2	$\frac{dy}{dx} = 5^{\sqrt{x}}(-2xe^{-x^2}) + (e^{-x^2})(5^{\sqrt{x}} \ln 5(\frac{1}{2}x^{-\frac{1}{2}}))$ (a) $= 5^{\sqrt{x}}e^{-x^2}(-2x + \frac{\ln 5}{2\sqrt{x}})$ $\frac{dy}{dx} = \ln(\sin \sqrt{x}) - \frac{1}{2}\ln(x^2 - 1)$ (b) $= \frac{\cos \sqrt{x}}{2\sqrt{x} \sin \sqrt{x}} - \frac{2x}{2(x^2 - 1)}$ $= \frac{\cot \sqrt{x}}{2\sqrt{x}} - \frac{x}{(x^2 - 1)}$	K1K1K1 J1 K1 K1K1K1 J1	Product rule + diff 1st correct + diff 2nd correct Law of log Diff ln + diff 1st function + diff 2nd function
Total		9 marks	
3(a)	$f(x) = 2x^3 - 12x^2 + 18x + 5$ $f'(x) = 6x^2 - 24x + 18$ $f'(x) = 0$ $6x^2 - 24x + 18 = 0$ $x^2 - 4x + 3 = 0$ $(x-3)(x-1) = 0$ $x = 3 \text{ or } x = 1$	K1 K1 K1	

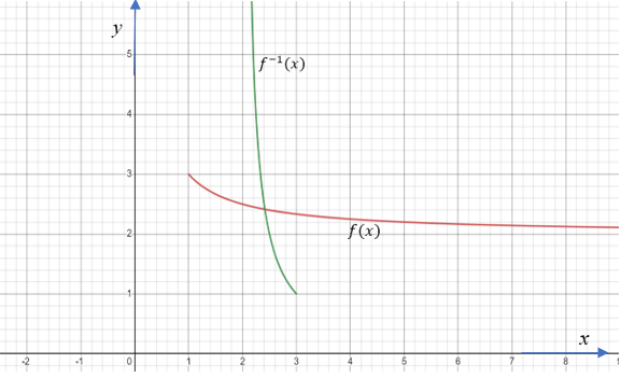
[illegible]

NO.	SOLUTION	MARKS	REMARKS
PART B			
1	$\frac{\sqrt{7} - xi}{1 - \sqrt{7}i} \times \frac{1 + \sqrt{7}i}{1 + \sqrt{7}i}$ $= \frac{\sqrt{7} + 7i - xi + \sqrt{7}x}{1 + 7}$ $= \frac{\sqrt{7} + \sqrt{7}x + (7 - x)i}{8}$ This is real number so imaginary part = 0 $\frac{7 - x}{8} = 0$ $\therefore x = 7$	K1 K1 K1 K1 J1	Conjugate Substitute i ² =1 Simplify Im(z) = 0
Total		5 marks	
2(a)	$\log_2 2x = \log_4 (x + 3)$ $\log_2 2x = \frac{\log_2 (x + 3)}{\log_2 4}$ $2 \log_2 2x = \log_2 (x + 3)$ $(2x)^2 = (x + 3)$ $4x^2 - x - 3 = 0$ $x = -\frac{3}{4}, x = 1$ $\therefore x = 1$	K1 K1 K1J1	Change base Compare using log base 2 Solve

NO.	SOLUTION	MARKS	REMARKS
2(b)	$\left \frac{1}{1-x} \right \geq 3$ $\frac{1}{1-x} \geq 3 \quad \text{OR} \quad \frac{1}{1-x} \leq -3$ $\frac{1}{1-3(1-x)} \geq 0 \quad \frac{1}{1+x} \leq 0$ $\frac{1-x}{3x-2} \geq 0 \quad \frac{4-3x}{1-x} \leq 0$ $\frac{1-x}{3x-2} \geq 0 \quad \frac{4-3x}{3x-4} \leq 0$ $\frac{1-x}{2} \geq 0 \quad \frac{x-1}{4} \leq 0$ $x \geq \frac{2}{3}, x < 1 \quad x \geq \frac{4}{3}, x > 1$  $\frac{2}{3} \leq x < 1$  $1 < x \leq \frac{4}{3}$  $\therefore \left[\frac{2}{3}, 1 \right) \cup \left(1, \frac{4}{3} \right]$	K1 K1 K1 K1 J1 K1 J1	Define modulus Equate denominator Simplify Method to solve Answer for each set Combine Final answer

NO.	SOLUTION	MARKS	REMARKS
	ALTERNATIVE: $\left \frac{1}{1-x} \right \geq 3$ $1 \geq 3 1-x , x \neq 1$ $ 1-x \leq \frac{1}{3}$ $-\frac{1}{3} \leq 1-x \leq \frac{1}{3}$ $-\frac{4}{3} \leq -x \leq -\frac{2}{3}$ $\frac{2}{3} \leq x \leq \frac{4}{3}$ $\therefore \left[\frac{2}{3}, 1 \right) \cup \left(1, \frac{4}{3} \right]$	K1 B1 K1 K1 K1 K1 J1	Cross multiply Condition Simplify Define modulus Make 'x' subject Solve
Total		11 marks	
3	(a) $R_{g(x)} = [2, \infty)$	B1	
	b) $fg(x) = \frac{2[x^2 + 2] + 3}{2[x^2 + 2] - 3}$ $= \frac{2x^2 + 4 + 3}{2x^2 + 4 - 3}$ $= \frac{2x^2 + 7}{2x^2 + 1}$	K1 K1 J1	

NO.	SOLUTION	MARKS	REMARKS
	c) $f(x) = \frac{2x+3}{2x-3} \quad f \circ f^{-1}(x) = x$ $\frac{2f^{-1}(x)+3}{2f^{-1}(x)-3} = x$ $2f^{-1}(x)+3 = x(2f^{-1}(x)-3)$ $f^{-1}(x) = \frac{-3x-3}{2-2x}$ $f^{-1}(x) = \frac{3x+3}{2x-2}$	K1 J1	
	d) $f(x) = f^{-1}(x)$ $\frac{2x+3}{2x-3} = \frac{3x+3}{2x-2}$ $(2x+3)(2x-2) = (3x+3)(2x-3)$ $2x^2 - 5x - 3 = 0$ $x = 3 \text{ and } x = -\frac{1}{2}$	K1 K1 J1	
Total		10 marks	

NO.	SOLUTION	MARKS	REMARKS
4	a) $y = 2$	B1	
	b) $R_{f(x)} = 2 < f(x) < 3$	B1	
	c) $f(x) = \frac{1}{x} + 2 \quad f \circ f^{-1}(x) = x$ $\frac{1}{f^{-1}(x)} + 2 = x$ $\frac{1}{f^{-1}(x)} = x - 2$ $f^{-1}(x) = \frac{1}{x - 2}$	K1 K1J1	
	c) $D_{f^{-1}(x)} = 2 < x < 3$	B1	
	$R_{f^{-1}(x)} = y > 1$	B1	
	e) 	R1 R1 R1	

NO.	SOLUTION	MARKS	REMARKS
Total		10 marks	
5(a)	$f(x) = \frac{2x-3}{(x-1)(x+3)}$ $(x-1)(x+3) \neq 0$ $x-1 \neq 0, \quad x+3 \neq 0$ $x \neq 1, \quad x \neq -3$ So, $D_f = (-\infty, 1) \cup (1, 3) \cup (3, \infty)$	B1	State domain of $f(x)$
5(b)	$f(x) = \frac{2x-3}{(x-1)(x+3)}$ x-intercept, let $f(x) = 0$ $\frac{2x-3}{(x-1)(x+3)} = 0$ $2x-3 = 0$ $x = \frac{3}{2}$ y-intercept, let $x = 0$	K1	Try to find both intercepts

NO.	SOLUTION	MARKS	REMARKS
	$f(0) = \frac{2(0)-3}{(0-1)(0+3)}$ $f(0) = \frac{-3}{-3} = 1$ $\therefore x\text{-intercept is at } \left(\frac{3}{2}, 0\right)$ $\therefore y\text{-intercept is at } (0, 1)$	J1 J1	Both correct
5(c)	At $x = 1$ $\lim_{x \rightarrow 1^+} \frac{2x-3}{(x-1)(x+3)} = -\infty$ $\lim_{x \rightarrow 1^-} \frac{2x-3}{(x-1)(x+3)} = \infty$ At $x = -3$ $\lim_{x \rightarrow -3^+} \frac{2x-3}{(x-1)(x+3)} = \infty$ $\lim_{x \rightarrow -3^-} \frac{2x-3}{(x-1)(x+3)} = -\infty$ Hence, $x = 1$ and $x = -3$ are vertical asymptotes.	K1 K1 J1	Try to prove asymptote at $x = 1$ Try to prove asymptote at $x = -3$ Conclusion

NO.	SOLUTION	MARKS	REMARKS
5(d)	$\lim_{x \rightarrow -\infty} \frac{2x-3}{(x-1)(x+3)}$ $= \lim_{x \rightarrow -\infty} \frac{2x-3}{x^2+2x-3}$ $= \lim_{x \rightarrow -\infty} \frac{\frac{2x-3}{x^2}}{\frac{x^2+2x-3}{x^2}}$ $= \lim_{x \rightarrow -\infty} \frac{\frac{2}{x} - \frac{3}{x^2}}{1 + \frac{2}{x} - \frac{3}{x^3}}$ $= \frac{0-0}{1+0-0} = 0$ $\lim_{x \rightarrow \infty} \frac{2x-3}{(x-1)(x+3)}$	K1 K1 K1	Divide by x^2 Simplify Use $\lim_{x \rightarrow \infty} \frac{\Re}{X} = 0$

NO.	SOLUTION	MARKS	REMARKS
	$= \lim_{x \rightarrow \infty} \frac{2x - 3}{x^2 + 2x - 3}$ $= \lim_{x \rightarrow \infty} \frac{\frac{2}{x} - \frac{3}{x^2}}{1 + \frac{2}{x} - \frac{3}{x^3}}$ $= \frac{0 - 0}{1 + 0 - 0} = 0$ Hence, $y = 0$ is horizontal asymptote.	J1	Conclusion
5(e)		R1 R1 R1	Shape Lable All correct

NO.	SOLUTION	MARKS	REMARKS
Total		14 marks	
6a)	$(c) \ f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} [\sqrt{1-3(x+h)} - \sqrt{1-3x}]$ $= \lim_{h \rightarrow 0} \frac{1}{h} \left[\sqrt{1-3x-3h} - \sqrt{1-3x} \times \frac{\sqrt{1-3x-3h} + \sqrt{1-3x}}{\sqrt{1-3x-3h} + \sqrt{1-3x}} \right]$ $= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{1-3x-3h-(1-3x)}{\sqrt{1-3x-3h} + \sqrt{1-3x}} \right]$ $= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{-3h}{\sqrt{1-3x-3h} + \sqrt{1-3x}} \right]$ $= \lim_{h \rightarrow 0} \left[\frac{-3}{\sqrt{1-3x-3h} + \sqrt{1-3x}} \right]$ $= -\frac{3}{2\sqrt{1-3x}}$	K1 K1 K1 J1	Multiply with conjugate Simplify
6b)			

NO.	SOLUTION	MARKS	REMARKS
	$x + t = xt$ $\frac{dx}{dt} + 1 = x + t \frac{dx}{dt}$ $(1 - t) \frac{dx}{dt} = x - 1$ $\frac{dx}{dt} = \frac{x - 1}{1 - t}$ $2ty - y^2 = 3$ $2t \frac{dy}{dt} + 2y - 2y \frac{dy}{dt} = 0$ $(y - t) \frac{dy}{dt} = y$ $\frac{dy}{dt} = \frac{y}{y - t}$	K1 J1 K1 J1	

NO.	SOLUTION	MARKS	REMARKS
	$x = 2,$ $2 + t = 2t$ $t = 2$	K1	Find t
	$2ty - y^2 = 3$ $2(2)y - y^2 = 3$ $y^2 - 4y + 3 = 0$ $(y - 3)(y - 1) = 0$ $y = 1, y = 3$	K1 J1	Find y
	$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{y}{1-t}}{\frac{y-t}{x-1}}$ $= \frac{y(1-t)}{(y-t)(x-1)}$	K1	Correct formula
	when $x = 2, t = 2, y = 1,$ $\frac{dy}{dx} = \frac{1(1-2)}{(1-2)(2-1)} = 1$	K1 J1	Substitute

NO.	SOLUTION	MARKS	REMARKS
	when $x = 2, t = 2, y = 3,$ $\frac{dy}{dx} = \frac{3(1-2)}{(3-2)(2-1)} = -3$	K1 J1	Substitute
Total		16 marks	
7(a)	$\frac{dV}{dt} = 6\pi$ $V = \pi h^2 \left(5 - \frac{1}{3}h \right) = \pi \left(5h^2 - \frac{h^3}{3} \right)$ $\frac{dV}{dh} = \pi(10h - h^2)$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{1}{\pi(10h - h^2)} \times 6\pi$ $= \frac{6}{10h - h^2}$ When $h = 4\text{cm},$ $\frac{dh}{dt} = \frac{6}{10(4) - 4^2} = \frac{1}{4} \text{ cm/s}$	K1 K1 K1J1	

NO.	SOLUTION	MARKS	REMARKS
7(b)	$r^2 + (8 - h)^2 = 8^2$	K1	
	$2r \frac{dr}{dt} + 2(8 - h) \left(-\frac{dh}{dt} \right) = 0$	K1	
	$r \frac{dr}{dt} - (8 - h) \frac{dh}{dt} = 0$		
	When $h = 4\text{cm}$,		
	$r^2 + (8 - 4)^2 = 8^2$		
	$r^2 + 16 = 64$	K1	
	$r^2 = 48$		
	$r = 4\sqrt{3}$ since $r > 0$		
	$4\sqrt{3} \frac{dr}{dt} - (8 - 4) \left(\frac{1}{4} \right) = 0$		
	$4\sqrt{3} \frac{dr}{dt} = 1$	K1	
	$\frac{dr}{dt} = \frac{1}{4\sqrt{3}} \text{ cm/s}$		
		J1	
Total		9 marks	